

Predicting Tax Refund or Tax Owed Outcomes Using Financial Activity

Patterns

Final Report / White Paper

Adhilingam Adiswaran

Bellevue University

DSC 680

October 2026

Business Problem and Research Questions

Many taxpayers do not know until late in the year whether federal withholding and estimated payments are aligned with eventual tax liability. An early classification of refund versus tax owed could support withholding review, cash-flow planning, and financial education. This project evaluates whether common financial and tax-related indicators can classify the likely direction of the outcome. It is an educational data-science prototype and is not intended to calculate an official tax return or provide tax advice.

The project addresses four research questions: (RQ1) Which financial and tax-related variables are most strongly associated with refund versus tax owed? (RQ2) How accurately can supervised classification models predict the outcome? (RQ3) Does a nonlinear Random Forest materially improve predictive performance over interpretable Logistic Regression? (RQ4) Does performance differ across income ranges or filing-status groups?

Background

Federal income-tax outcomes depend on the relationship between estimated liability and taxes already paid through withholding and estimated payments. Liability itself is influenced by filing status, income, deductions, adjustments, and credits. These relationships make withholding, estimated payments, and liability-related variables natural candidates for an early-warning classification model (Internal Revenue Service [IRS], n.d.-a).

Because linked banking and tax-return records are highly sensitive and difficult to obtain ethically for a classroom project, this study uses synthetic profiles. Public IRS Statistics of Income concepts were used to guide plausible variables and relationships without claiming access to confidential taxpayer records (IRS, n.d.-b; IRS, 2024).

Data and Preparation

The analysis uses 5,000 synthetic individual records generated with a fixed random seed of 680. Variables include annual wages, other income, adjusted gross income, filing status, dependents, deductions, credits, federal withholding, estimated tax payments, annual expenses, savings behavior, and income consistency. The binary target is Refund or Tax Owed. No names, Social Security numbers, account numbers, addresses, or real transactions are included.

Feature engineering created withholding-to-income, estimated-payment-to-income, income-to-expense, savings-rate, and total-tax-payments-to-estimated-liability measures. Filing status was one-hot encoded and continuous variables were standardized where required. Direct outcome fields such as a final refund amount or balance due were excluded to reduce target leakage. The notebook also performs explicit range, missing-value, and duplicate checks so the workflow can be adapted to external data.

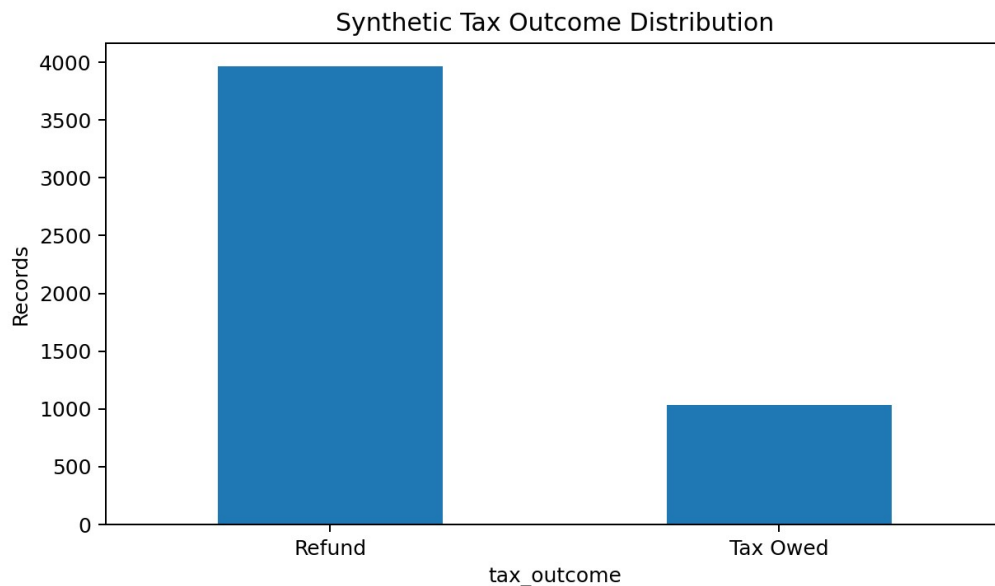


Figure 1. Distribution of the 5,000 synthetic tax outcomes.

Methods

Three supervised classifiers were evaluated. Logistic Regression provides an interpretable baseline, a Decision Tree captures readable nonlinear thresholds, and Random Forest captures nonlinear interactions while reducing the instability of a single tree. The dataset was split into 75% training and 25% held-out testing sets using stratification. Five-fold stratified cross-validation was performed on the training data.

Evaluation included accuracy, precision, recall, F1 score, ROC-AUC, confusion matrices, permutation importance, and subgroup performance. Multiple metrics were used because the generated target is imbalanced: 79.3% of records are Refund and 20.7% are Tax Owed. Subgroup results were calculated by income range and filing status to identify potential generalization and fairness concerns.

Analysis and Results

Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	87.84%	0.954	0.890	0.921	0.945
Decision Tree	85.20%	0.958	0.851	0.901	0.929
Random Forest	90.16%	0.944	0.931	0.938	0.940

Random Forest produced the highest held-out accuracy (90.16%) and F1 score (0.938), while Logistic Regression produced the highest ROC-AUC (0.945). Five-fold mean ROC-AUC values were 0.942 for Logistic Regression, 0.918 for the Decision Tree, and 0.936 for Random Forest. Therefore, the nonlinear ensemble improved accuracy and F1, but did not dominate the interpretable baseline on every metric.

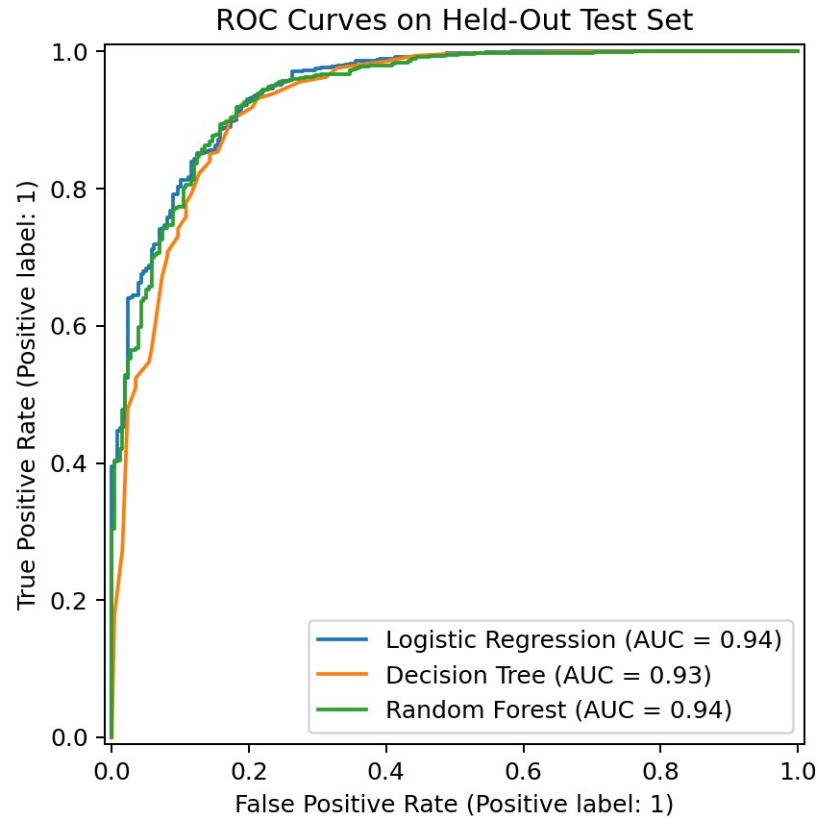


Figure 2. ROC curves for the three classifiers on the held-out test set.

For Random Forest, the confusion matrix contained 204 correctly identified Tax Owed records and 923 correctly identified Refund records. Fifty-five Tax Owed records were incorrectly classified as Refund, while 68 Refund records were incorrectly classified as Tax Owed. This error pattern matters because an incorrect refund prediction could leave a user financially unprepared for a balance due.

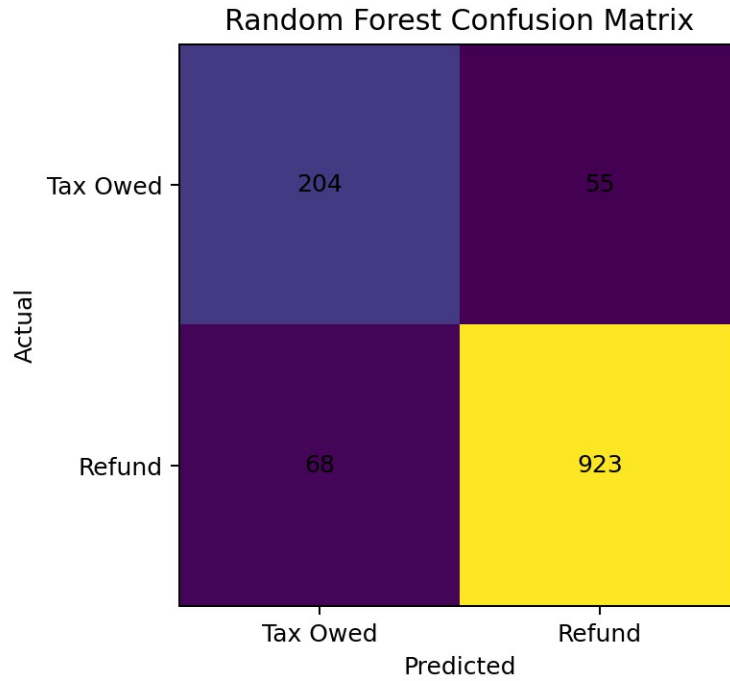


Figure 3. Random Forest confusion matrix.

Permutation importance identified `tax_payments_to_liability` as the dominant predictor, followed by `withholding_to_income` and `federal_withholding`. This answers RQ1: in the synthetic design, the strongest signal is the relationship between taxes already paid and estimated liability. Feature importance should be interpreted as association within this generated experiment, not as proof of causation.

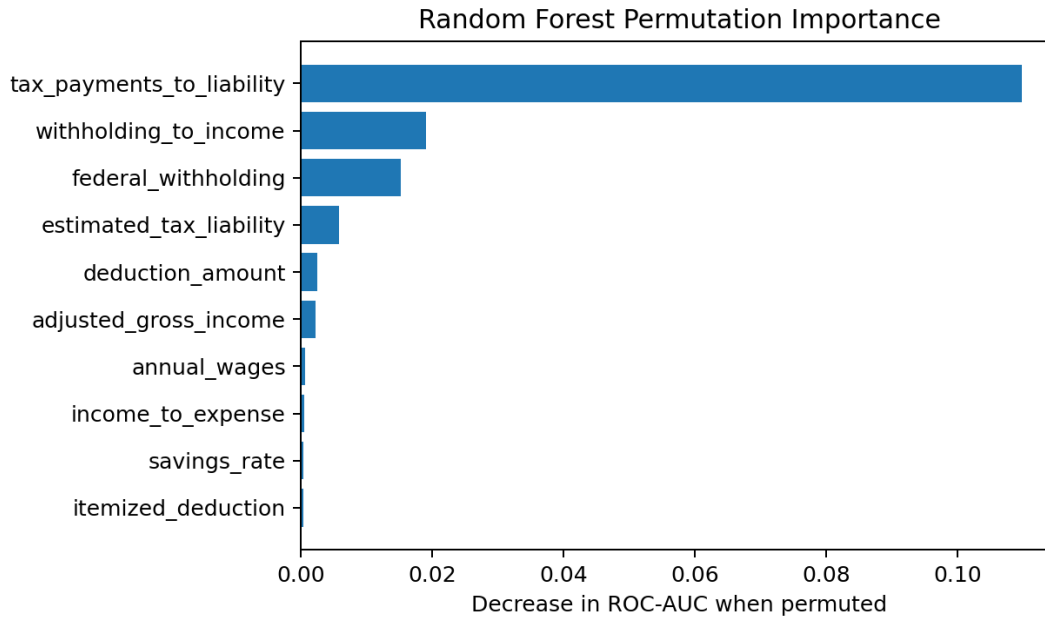


Figure 4. Random Forest permutation importance.

Subgroup Analysis and Research Question Answers

Performance varied across subgroups. Random Forest accuracy ranged from 84.08% for the \$100,000-\$200,000 income group to 92.75% below \$50,000. By filing status, accuracy ranged from 87.31% for Single to 93.22% for Head of Household. The \$200,000+ test subgroup contained only 56 records and had zero correctly recalled Refund cases, so that estimate is unstable and should not be generalized.

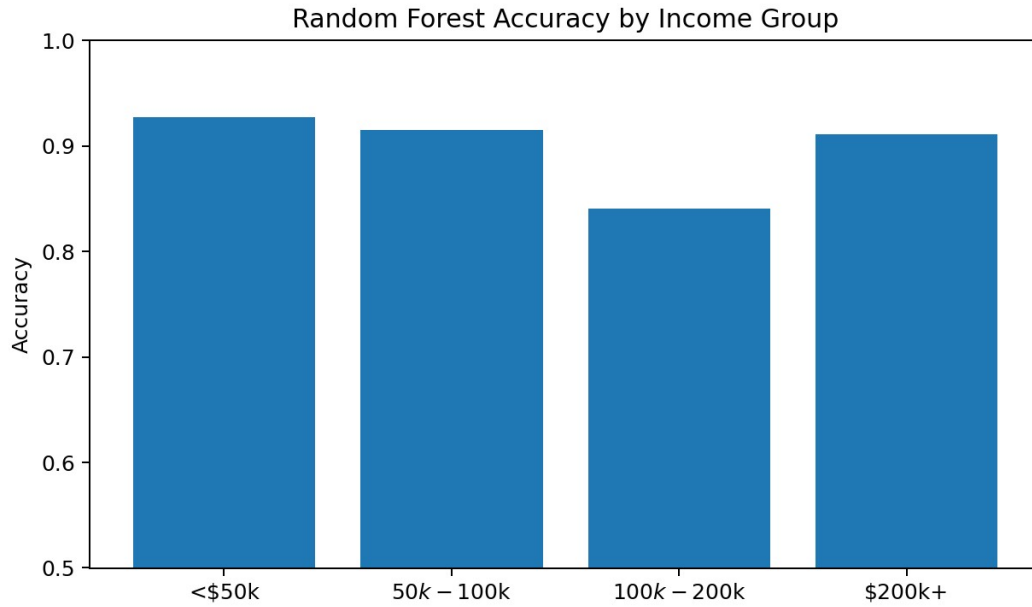


Figure 5. Random Forest accuracy by synthetic income group.

RQ2 is answered by the held-out results: all three models demonstrated useful discrimination in the generated data, with Random Forest reaching 90.16% accuracy. RQ3 is answered with a qualified no: Random Forest materially improved accuracy and F1 relative to Logistic Regression, but Logistic Regression retained a slightly higher ROC-AUC and offers easier interpretation. RQ4 is answered yes: performance differed across income and filing-status groups, although the synthetic data and small subgroup sizes do not support population-level fairness conclusions.

Ethical Assessment and Limitations

The primary ethical concerns are privacy, subgroup bias, synthetic-data validity, target leakage, overinterpretation of feature importance, and misuse of predictions as tax advice. Privacy risk is reduced because the project uses synthetic records with no direct identifiers. Bias risk remains because subgroup performance varies, and synthetic generation may not reflect real taxpayers equally well. A real implementation would require data minimization, access controls,

retention limits, subgroup monitoring, model documentation, and explicit user-facing uncertainty.

The largest limitation is external validity. Results from synthetic profiles cannot establish real-world performance. The simplified liability logic omits many tax-code provisions, tax rules change over time, and the strong payment-to-liability feature is partly expected because the synthetic target is constructed from related payment logic. Any production extension would require validation on legally obtained, de-identified real data and tax-year-specific review.

Conclusion and Recommendations

This project demonstrates that financial and tax-related indicators can classify refund-versus-tax-owed status with strong performance in a controlled synthetic experiment. Random Forest achieved the best accuracy and F1, while Logistic Regression remained highly competitive and slightly stronger on ROC-AUC. The central domain finding is that tax payments relative to estimated liability provide the strongest predictive signal.

The model should be positioned only as an educational screening tool. A future implementation could display a probability band and key contributing factors, prompt users to review withholding when the probability of owing tax is elevated, and route users to official IRS tools or a qualified tax professional. Before production use, the system would need real-world validation, annual tax-rule review, privacy controls, fairness monitoring, model governance, and human review for ambiguous cases.

Code and Reproducibility

The accompanying Jupyter notebook, `DSC680_Tax_Outcome_Prediction_Notebook_Executed.ipynb`, contains synthetic data

generation, data-quality checks, feature engineering, model training, five-fold cross-validation, evaluation, subgroup analysis, and figure generation. It uses a fixed random seed and standard Python libraries including pandas, NumPy, Matplotlib, and scikit-learn.

References

Internal Revenue Service. (n.d.-a). Tax Withholding Estimator FAQs.

<https://www.irs.gov/individuals/tax-withholding-estimator-faqs>

Internal Revenue Service. (n.d.-b). SOI tax stats - Individual income tax returns complete report

(Publication 1304). <https://www.irs.gov/statistics/soi-tax-stats-individual-income-tax-returns-complete-report-publication-1304>

Internal Revenue Service. (2024). Statistics of Income - 2022 individual income tax returns, line item estimates (Publication 4801). <https://www.irs.gov/pub/irs-pdf/p4801.pdf>

Tabassi, E. (2023). Artificial Intelligence Risk Management Framework (AI RMF 1.0) (NIST AI 100-1). National Institute of Standards and Technology.

<https://doi.org/10.6028/NIST.AI.100-1>